

Chapter One: Plant Biology

Plants manufacture their own food through a process that captures sunlight. Chlorophyll in the chloroplasts absorbs photons and drives the conversion of carbon dioxide and water into glucose and oxygen. The light dependent reactions occur in the thylakoid membrane, while the Calvin cycle fixes carbon in the stroma. A single leaf can host roughly five hundred thousand chloroplasts per square millimetre.

Chapter Two: European History

The Treaty of Versailles was signed on the twenty eighth of June nineteen nineteen in the Hall of Mirrors. It imposed reparations of one hundred thirty two billion gold marks on Germany and redrew the borders of central Europe. Article two hundred thirty one assigned responsibility for the war, a clause that became deeply resented.

Chapter Three: Algorithms

Binary search locates a value in a sorted array by repeatedly halving the interval under consideration. Because the search space shrinks by half on every comparison, the running time is logarithmic. An array of one million elements needs at most twenty comparisons. The array must already be sorted, which is the precondition people most often forget.